PES UNIVERSITY

Department of Computer Science and Engineering

UE19CS251: Design and Analysis of Algorithms (4-0-0-4-4)

Unit 4:

Question Bank-Space and Time Tradeoffs and Greedy method

1. What is the concept of Time and Space Trade-off? List the different techniques that helps to improve the time efficiency.

2. Assuming that the set of possible list values is {*a, b, c, d*}, sort the following list in alphabetical order by the distribution-counting algorithm: *b, c, d, c, b, a, a, b*

3.Define Input enhancement and Prestructuring.

4. Explain the method for making the “Shift Table” used by Horspool and Boyer-Moore algorithm.

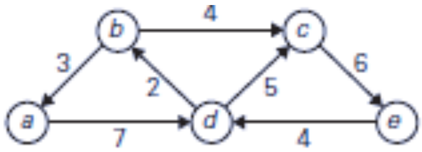
5. Write Horspool Algorithm for String matching. Trace the algorithm to find the pattern “ELECTION” in the text “EDUCATION_ONLY_HELPS_IN_SELECTION”.

6. In Boyer-Moore algorithm, explain when to use bad-symbol table and good-suffix table with suitable illustrations.

7. a) Construct the shift table for the following gene segment of your chromosome 10: TCCTATTCTT
b) Apply Horspool’s algorithm to locate the above pattern in the following DNA sequence: TTATAGATCTCGTATTCTTTTATAGATCTCCTATTCTT

8. Apply Boyers-Moore algorithm to search for the pattern AT_THAT in the text WHICH_FINALLY_HALTS.__AT_THAT.

9. Solve the following instances of the single-source shortest-paths problem with vertex a as the source:



10. Compare fixed-length encoding, variable-length encoding and prefix-free code.

11. For data transmission purposes, it is often desirable to have a code with a minimum variance of the codeword lengths (among codes of the same average length). Compute the average and variance of the codeword length in two Huffman codes that result from a different tie breaking during a Huffman code construction for the following data:

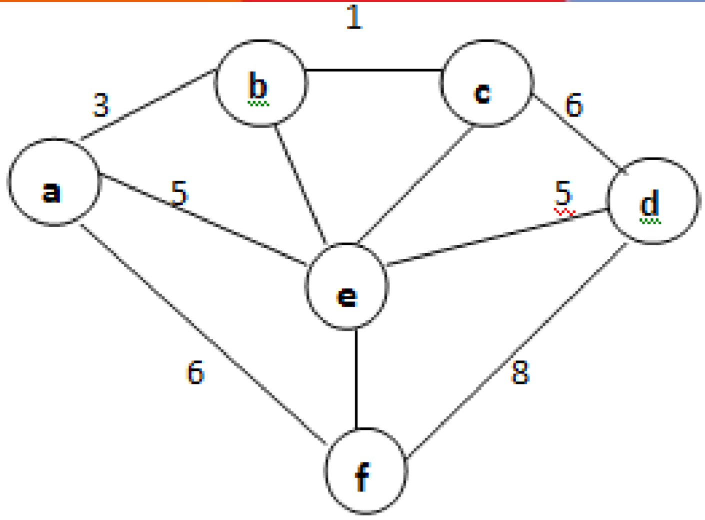
symbol	A	B	C	D	E
probability	0.1	0.1	0.2	0.2	0.4

12. Differentiate Greedy method with Dynamic programming?

13. Explain the comparison counting algorithm. What is the efficiency of this algorithm?

14. Explain the algorithm for computing the shift table entries.

15. Using Prim's algorithm, determine minimum cost spanning tree for the weighted graph shown below.



16. i) Write Kruskal’s algorithm to find the minimum cost spanning tree.

ii) Using Kruskal’s algorithm, determine minimum cost spanning tree for the weighted graph shown in the previous question.

17. Write a note on Huffman Trees anf Huffman code.

18. Construct a Huffman code for the following data:

Character :	A	B	C	D	-
Probability:	0.4	0.1	0.2	0.15	0.15

19. Explain Greedy Technique.

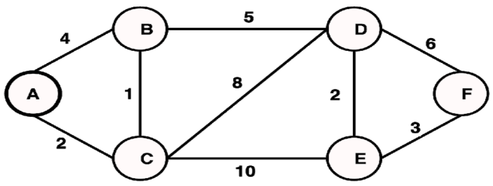
20. Write and explain Dijkstra’s algorithm for single source shortest distance problem. Also derive the time complexity of it.

21. What is the efficiency of Kruskal’s and prims algorithm? Compare

22. Explain how find and union operations of Kruskal’s algorithm is implemented.

23. Write the control abstraction for the subset paradigm.

24. Apply Dijkstra's algorithm to find single source shortest paths for the following graph taking vertex 'A' as source.



25. i) Write the control abstraction for the subset paradigm.
    ii) Define minimum cost spanning tree
    iii) Define the following functions:
        a) makeset(x),
        b) find(x),
        c) union(x,y).

26. List down the methods to implement prestructuring and explain them briefly.

27. Write the pseudocode for Distribution counting sort and analyse its Time efficiency.

28. Is the distribution-counting algorithm stable?

29. Explain with examples the purpose of bad-symbol table and good-suffix table.

30. Explain Greedy technique using change-making problem.

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